

ARIN 340 Unit 3 Assignment Worksheet

(This worksheet will be submitted at the end of Unit 3.)

Unit 3 Assignment Directions:

1. You will complete the seven chapters in the [Prompt Engineering For Everybody With ChatGPT and GPT4](#) tutorial.
2. After each chapter, you will have an exercise or tasks to complete within this document. **NOTE: Your prompts and output for this document should NOT simply replicate the steps or tasks within the tutorial. Use new prompts to generate new output.** In tasks when prompts are required, include the prompts and output below, and respond to any questions.
3. When prompts and responses are required, cut and pasted text is preferred. You may use screenshots or snips, but the text must be clear and legible. Any generated images will be screenshots or snips.
4. After you have completed the tutorial and the exercises and tasks here, you can submit this document to the Unit 3 Assignment folder in LEO.

This assignment is worth 140 points total, and you can receive up to 20 points per chapter entry.

Gheorghiu, A. (2023). *Prompt engineering for everyone with CHATGPT and GPT-4*. O'Reilly Media.

Chapter 1: Introduction to Prompt Engineering

Task: Write a paragraph summary of key concepts and definitions in prompt engineering.

Prompt engineering is the creation of prompts for LLMs to produce desired outputs. Prompts are the commands given to AI Large Language Models (LLMs). Large Language Models are AI models that understand, recognize, process, and produce natural language. Prompts are communicated to the LLMs through natural human language that the LLM can recognize. Prompt Engineering involves understanding how language models interpret prompt instructions and how to guide them toward useful, accurate, and context-aware responses. Chapter 1 shows that prompt engineering is a foundational skill for interacting with AI tools like ChatGPT, Bard, or Bing AI, and is increasingly used in industries across the board. It is redefining how we work. Some of the key concepts include the structure of prompts, the importance of clarity and specificity, and the refining of prompts to improve results. One of the biggest concepts of Prompt Engineering is that your prompts should be detailed, concise, and worded to achieve the best results. Chapter 1 also introduces ethical considerations, such as responsible AI use and bias awareness, and highlights the potential of prompt engineering in enhancing productivity and creativity. There are some key terms used in Prompt Engineering such as token, generative AI, AGI, and GPT. A token is a chunk of text that is given to a LLM to be processed. This can be a single character or a combination of words. Generative AI is AI that creates new content from the data it is given. AGI is Artificial General Intelligence. This is AI thinking like a human and being able to reason and think like humans. This type of AI does not exist today. GPT is a Generative Pre-Trained Transformer. This means they are LLMs that are trying to correctly predict the next word in a phrase, which is communicated in a human like response. These are some of the basic concepts of Prompt Engineering, though there are more.

Chapter 2: Basics of Prompt Engineering

Task: Create a set of simple prompts and demonstrate an iterative process for refining them (at least two iterations after the initial prompt).

Respond to the following questions:

- How did the iterations produce better results?
- How might you have changed your prompts to produce better results

Initial Prompt–“Create a list of the best hybrid cars?”

Initial Response–“ Here’s a refined and up-to-date list of the **best hybrid cars of 2025**, based on top automotive reviews and expert rankings:

Best Overall Hybrid Cars (Sedans)

Rank	Model	Highlights
①	2025 Toyota Camry Hybrid	Rated the best hybrid car of 2025 by both Kelley Blue Book and U.S. News for its balance of performance, features, and fuel efficiency (~44–53 MPG city / 43–50 MPG highway). Kbb.com U.S. News Cars
②	2025 Honda Civic Hybrid	Named “Top Rated Car” by Edmunds for excellent fuel economy (~50 city / 47 highway MPG), a refined interior, and lively performance. AP News +1U.S. News Cars
③	2025 Hyundai Elantra Hybrid	Best fuel economy among mainstream hybrids (~54 MPG combined), strong value, low starting price. Kbb.com U.S. News Cars

Other notable sedans:

- Honda Accord Hybrid: Smooth ride, spacious, solid fuel economy (~46–51 city / 41–44 highway MPG); tied for second on U.S. News. [U.S. News Cars](#)[+1](#)
- Toyota Prius: Continues as the iconic benchmark for hybrids with stellar efficiency (~57/56 MPG city/highway); retains strong reputation. [Phoenix Transportation San Francisco](#)[MotorTrend](#)[DAX StreetAutosoto](#)

Best Hybrid SUVs

Mainstream Hybrid SUVs

- 2025 Hyundai Tucson Hybrid – Topped U.S. News SUV rankings with ~37–38 MPG, great performance, tech, and comfort. [U.S. News Cars](#)
- 2025 Kia Sportage Hybrid – Excellent value, roomy, enjoyable drive, good fuel economy (~38–44 MPG). [U.S. News Cars](#)[Car and Driver](#)[AOL](#)
- 2025 Ford Escape Hybrid – Compact, efficient (~42 city / 37 highway MPG), versatile and well-equipped. [Phoenix Transportation San Francisco](#)[Autofreak](#)[Autosoto](#)
- 2025 Honda CR-V Hybrid – Reliable and spacious SUV with ~40 MPG combined. [Phoenix Transportation San Francisco](#)[Road & Track](#)

Luxury Hybrid SUVs

- 2025 Lexus RX Hybrid – Top-rated luxury hybrid SUV per U.S. News; elegant with solid economy. [U.S. News Cars](#)
- 2025 Lexus UX 300h – A subcompact luxury SUV that delivers ~43 MPG combined, updated powertrain, and strong safety features. [MarketWatch](#)[Road & Track](#)
- 2025 Lexus RX 500h – High-performance luxury hybrid (~37 MPG), plush interior with sporty performance. [Phoenix Transportation San Francisco](#)[Autofreak](#)

Special Mentions & Ratings

- 2025 Toyota Prius Prime (Plug-In Hybrid) – Offers ~54 MPG plus ~40 miles of electric-only range. Great for commuters. [Phoenix Transportation San Francisco](#)
- 2025 Toyota RAV4 Hybrid / New RAV4 electrified lineup – The RAV4 is transitioning to a full hybrid and plug-in hybrid lineup in 2025–2026. [The Sun](#)[Wikipedia](#)
- Edmunds Top Rated SUV of 2025: Toyota Grand Highlander Hybrid—roomy three-row SUV that achieves ~36 MPG combined. [AP News](#)

Quick Recap: Top Picks

Compact/Mid-size Hybrid Sedans

- Toyota Camry Hybrid
- Honda Civic Hybrid
- Hyundai Elantra Hybrid
- Honda Accord Hybrid
- Toyota Prius

Hybrid SUVs

- Hyundai Tucson Hybrid
- Kia Sportage Hybrid

- Ford Escape Hybrid
- Honda CR-V Hybrid

Luxury Hybrid SUVs

- Lexus RX Hybrid
- Lexus UX 300h
- Lexus RX 500h

First Iteration-“Create a list of the best hybrid sedans that have the best gas efficiency?”

First Iteration Response-

“ Here’s a carefully curated list of the **most fuel-efficient hybrid sedans of 2025**, based on official EPA ratings and multiple trusted automotive sources:

Top Hybrid Sedans by Combined Fuel Economy (2025)

1. Toyota Prius

- Combined mileage: ~57 MPG, with AWD versions offering around 54 MPG. Still the overall king of fuel efficiency among hybrids.
[EdmundsForbesRoad & Track](#)

2. Hyundai Elantra Hybrid

- Combined mileage: up to 54 MPG (EPA-estimated). Its Blue trim specifically achieves 51 city / 58 highway MPG.
[EdmundsTop SpeedAP News](#)

3. Kia Niro (Hybrid)

- Combined mileage: 53 MPG. Though classified as a compact crossover, its efficiency rivals the top sedan offerings.
[EdmundsForbes](#)

4. Toyota Camry Hybrid

- Combined mileage: ~51 MPG in its base LE trim; AWD models around 50 MPG.
[EdmundsMarketWatchForbes](#)

5. Toyota Corolla Hybrid

- Combined mileage: approximately 50 MPG. Some estimates show 53 city / 46 highway MPG, averaging around 50 MPG.
[EdmundsAutoblogForbes](#)

6. Honda Civic Hybrid

- Combined mileage: around 49 MPG. Delivers strong performance (0–60 in ~6.6 seconds) while maintaining exceptional efficiency.
[EdmundsMotor1.comAP News](#)

7. Honda Accord Hybrid

- Combined mileage: about 48 MPG. Offers a spacious cabin with solid fuel economy.
[EdmundsMotor1.comSlashGear](#)

8. Hyundai Sonata Hybrid

- Combined mileage: roughly 47 MPG. Delivers sleek styling, a long range, and respectable efficiency for a midsize sedan.
[EdmundsAutoblogSlashGear](#)

9. Lexus ES 300h

- Combined mileage: ~44 MPG. A luxury pick that blends refinement with decent efficiency.
[ForbesSlashGear](#)

Summary Table: Fuel Efficiency Leaders

Rank	Model	Combined MPG
1	Toyota Prius	~57 MPG
2	Hyundai Elantra Hybrid	~54 MPG
3	Kia Niro (Hybrid)	~53 MPG
4	Toyota Camry Hybrid	~51 MPG
5	Toyota Corolla Hybrid	~50 MPG
6	Honda Civic Hybrid	~49 MPG
7	Honda Accord Hybrid	~48 MPG
8	Hyundai Sonata Hybrid	~47 MPG
9	Lexus ES 300h	~44 MPG

★ Highlights & Recommendations

- Ultimate efficiency: If your top priority is the highest MPG, the Toyota Prius is still unmatched.
- Efficient and stylish: The Hyundai Elantra Hybrid delivers near-top efficiency with modern design and great value.

- SUV alternative: The Kia Niro Hybrid offers crossover practicality with sedan-level fuel economy.
- Spacious hybrid: The Toyota Camry Hybrid strikes a great balance between efficiency and comfort (AWD available).
- Performance plus efficiency: The Honda Civic Hybrid offers strong acceleration alongside high MPG.
- Luxury hybrid option: The Lexus ES 300h brings premium features with respectable fuel economy.

Second Iteration-“Create a list of the best hybrid sedans that have the best gas efficiency rating and were built in America. After completing the list, re-rank the list by the sedan's safety rating.”

Here's your refined list:

1. Best Hybrid Sedans Built in America (by Fuel Efficiency)

Based on EPA fuel economy ratings and confirmed U.S. assembly:

A. 2025 Toyota Camry Hybrid (Built in the U.S.)

- EPA Fuel Efficiency:
 - Up to 51 MPG combined for the LE FWD (approximately 53 city / 50 highway)
[MotorTrendGreen Toyota](#)
 - Higher trims (SE/XLE/XSE) offer ~47 MPG combined; AWD versions range from ~50 to 44 MPG combined depending on trim
[MotorTrendGreen Toyota](#)
- Manufacturing: Assembled at Toyota Motor Manufacturing Kentucky (TMMK) in Georgetown, KY.
[MarketWatchWikipedia](#)

B. 2025 Honda Accord Hybrid (Built in the U.S.)

- EPA Fuel Efficiency:
 - Up to 48 MPG combined for the EX-L Hybrid (51 city / 44 highway)
[Kbb.comHeadquarter HondaKen Ganley Honda](#)
 - Other trims (Sport, Sport-L, Touring) average ~44 MPG combined (46 city / 41 highway)
[Ken Ganley HondaPatty Peck Honda](#)
 - Manufacturing: Honda has shifted Accord production to its Indiana plant (though not yet fully confirmed, production is moving there by 2025)
[Wikipedia](#)
-

2. Re-Ranked by Safety Ratings

Let's rank these two by their safety performance based on available crash-test data:

A. Toyota Camry Hybrid (2025)

- NHTSA: Earned a 5-Star Overall Safety Rating
[YouTube](#)
- IIHS: Achieved a Good (the highest) rating in side crash testing on occupant protection, indicating a strong safety structure
[IIHS](#)

B. Honda Accord Hybrid (2025)

- NHTSA Crash Rating: Specific data not found here, but frontal crash test footage indicates 5-Star ratings for driver and passenger in frontal full-overlap test
[YouTube](#)
- IIHS Safety: Not directly referenced for this hybrid, though the Accord generally performs strongly. Front crash prevention ("Collision Mitigation Braking System") earns an "A" (Acceptable) rating in pedestrian tests
[IIHS](#)

3. Final Rankings

By Fuel Efficiency

1. 2025 Toyota Camry Hybrid – ~51 MPG combined
2. 2025 Honda Accord Hybrid – ~48 MPG combined

By Safety Rating

1. 2025 Toyota Camry Hybrid
 - 5-Star NHTSA overall
 - IIHS "Good" side crash protection
2. 2025 Honda Accord Hybrid
 - Indicated 5-Star frontal NHTSA ratings via test footage
 - Acceptable IIHS pedestrian crash prevention

Summary Table

Rank	Model	MPG (Combined)	Built In U.S.?	Safety Highlights
1	2025 Toyota Camry Hybrid	~51 MPG	Yes – TMMK, Kentucky	5-Star NHTSA overall; IIHS Good in side crash

Rank	Model	MPG (Combined)	Built In U.S.?	Safety Highlights
2	2025 Honda Accord Hybrid	~48 MPG	Yes – (moving to Indiana)	Frontal 5-Star NHTSA; IIHS “A” in pedestrian front crash prevention

Key Takeaways

- Toyota Camry Hybrid leads both in fuel efficiency and overall safety performance it has documented.
- Honda Accord Hybrid remains very efficient and appears safe, though we lack comprehensive IIHS crash data specific to the 2025 hybrid model.

The iterations definitely focused the answers of my questions. I first asked for ChatGPT to just list the best hybrid cars. This gave me a combination of hybrid sedans, trucks, and suvs. They were all ranked by what is considered the best in general. My first iteration I changed the question to the best hybrid sedans instead of saying cars. I also added that I wanted the list to be based on the best cost efficiency. This changed my list even further. The list only contained sedans this time and they were ranked based on the cost efficiency. While most of the sedans were still on the initial prompt response, the order of the best shifted because the list was based on cost efficiency this time. My 2nd iteration I asked the same as the 1st iteration, but added I wanted the hybrid sedan to be built in America. I also then nested in another question by asking it to re-rank my list based off of the cars safety rating. Each iteration I asked gave me a more detailed response based off of me giving a more detailed prompt. I think there is much more I could have asked of the prompt. One, I could have asked for multiple lists to be made by multiple experts. For example a mechanics recommendation compared to a safety inspectors recommendation. I also think more details for what I want in a car would be useful too. Instead of gas efficiency I might add I want the hybrid to average 55mpg. I could also add a specific American brand I want to buy from or a certain price that I am aiming for.

Chapter 3: Advanced Prompt Engineering Techniques

Tasks:

1. Develop a complex prompt using advanced techniques, such as multilingual, multimodal, or chaos prompting.

I tried using the chaos prompting technique. I first asked for 5 names. I then asked to create a character background for each name. Finally, I asked for the 5 characters to be used in a one page short witty story about the workplace. This prompting did work. It created a 1 page story that was pretty adequate. I think the story could use more detail and changes in wording. I also noticed ChatGPT wanted to add all characters names and characteristics as much as possible. This took away from the story because it seemed the names and characteristics were just forced in because it had too. I think this shows that with chaos prompting you need to know how many variables you want or how many will work. You also might need to know the length or detail of output you need. An example is 5 characters is probably too many for a 1 page story. Overall, ChatGPT did a great job of creating a workable story basically from the start of name 5 names.

You said:

Initial Prompt-“Make a list of 5 names”

ChatGPT said:

Sure! Here are five names:

1. Sophia
2. James
3. Aiden
4. Isabella
5. Marcus

Would you like me to make them more themed (e.g., fantasy, modern, unique, cultural)?

You said:

Prompt 2-“Create a character background for each name”

ChatGPT said:

Got it! Here are character backgrounds for each of the five names I listed:

1. Sophia

- Age: 27
- Background: Sophia grew up in a bustling coastal city, the daughter of a shipwright and a teacher. She’s always been curious, with a love for books and ancient history. Her fascination with myths and legends led her to become an archaeologist, unearthing forgotten stories of civilizations long gone.

- Traits: Intelligent, determined, and empathetic.
 - Conflict: Sophia struggles with balancing her passion for discovery with the ethical dilemmas of removing artifacts from their resting places.
-

2. James

- Age: 34
 - Background: James was once a decorated soldier who left the military after witnessing the devastation of war. Now he works as a private investigator in a crime-ridden metropolis, trying to bring justice where the system has failed.
 - Traits: Stoic, loyal, and resourceful.
 - Conflict: Haunted by his past decisions on the battlefield, James often questions whether his pursuit of justice is driven by redemption or revenge.
-

3. Aiden

- Age: 19
 - Background: Aiden is a talented street musician who grew up in foster care, always moving from one place to another. Music became his anchor, his way of expressing the emotions he never felt safe speaking aloud.
 - Traits: Charismatic, creative, and restless.
 - Conflict: Aiden dreams of making it big, but fears losing the raw authenticity that makes his music resonate with others.
-

4. Isabella

- Age: 42
 - Background: Isabella is a skilled diplomat, raised in a family of politicians. She's traveled across continents, brokering peace agreements and navigating delicate alliances. Her elegant demeanor hides the fact that she's constantly under immense pressure.
 - Traits: Strategic, compassionate, and persuasive.
 - Conflict: Isabella secretly longs for a simpler life, away from political games, but fears abandoning her duty when so many depend on her.
-

5. Marcus

- Age: 29

- Background: Marcus is an engineer and inventor in a steampunk-inspired city powered by gears and steam. He grew up in poverty but used his ingenuity to climb into the ranks of respected creators. His inventions, however, have caught the eye of both powerful corporations and rebels who want to use them for their own purposes.
 - Traits: Brilliant, cautious, and inventive.
 - Conflict: Torn between selling his inventions to secure a future for his family or risking everything to help a revolution he quietly believes in.
-

Would you like me to expand these into full character sheets (appearance, skills, motivations, flaws, allies, enemies) for roleplay/storytelling use?

You said:

Prompt 3—"Use these 5 characters to write a witty short story about the workplace"

ChatGPT said:

Here's a witty short story starring Sophia, James, Aiden, Isabella, and Marcus—all trapped in the same workplace setting:

"The Office of Unlikely Talent"

The *Department of Extraordinary Affairs* was not your average workplace. Tucked inside a grimy downtown office building with flickering fluorescent lights and a printer that sounded like it was actively crying, five very different people somehow ended up working there.

Sophia adjusted her glasses and spread out a stack of ancient-looking scrolls on the breakroom table. "These are 14th-century manuscripts," she said, glaring at the coffee ring James had just set his mug on. "Do you *mind*?"

James shrugged, leaning back in his chair like the ex-soldier-turned-investigator he was. "It's a workplace. Papers are meant to get coffee stains. Builds character."

"*You* build character," muttered Aiden, who was strumming his guitar in the corner instead of actually working. He was technically "Head of Morale," though most of his morale-boosting came in the form of loud ballads about heartbreak.

"Head of Morale," Isabella repeated, pacing with her phone pressed to her ear. "Do you realize how absurd this organization looks when our *official* morale strategy is a nineteen-year-old with a guitar and no health insurance?"

"That's ageist," Aiden called. "And besides, my music is cheaper than therapy."

Meanwhile, Marcus was fiddling with the office coffee machine. Sparks flew out of it, followed by a puff of suspicious black smoke.

“Relax,” he said, waving his soot-covered hands. “It’s just a prototype. Soon, this machine will brew *three cups of coffee at once*.”

“It already does that,” Sophia said, deadpan.

“Yes,” Marcus replied, tightening a gear, “but mine also hums Beethoven’s Fifth while it brews.”

James sighed, rubbing his temples. “And I thought trench warfare was chaotic.”

By noon, the team was supposed to present a quarterly report to the *Board of Directors*. Instead, Isabella was still negotiating a ceasefire between Sophia and James over the proper storage of ancient manuscripts, Aiden was writing a protest song about office dress codes, and Marcus was frantically chasing down his runaway coffee machine, which had grown legs.

The meeting did not go well.

But somehow, through Sophia’s passionate storytelling, James’ no-nonsense grit, Aiden’s distracting charm, Isabella’s silver-tongued diplomacy, and Marcus’ wild inventions, the Board ended up giving the department a bigger budget.

No one was quite sure why.

As they filed out of the conference room, Isabella sighed.

“You’re all disasters,” she said. “But apparently, you’re *my* disasters.”

“Great,” Aiden grinned. “Does this mean I can expense a new guitar?”

James muttered something about needing stronger coffee. Sophia clutched her manuscripts protectively. Marcus sketched plans for a coffee machine that wouldn’t attack its owner.

And thus, another day at the Department of Extraordinary Affairs came to a close—proof that sometimes, the strangest workplace teams are the ones that accidentally work best.

1. If you have any experience in programming, demonstrate its application in writing code with AI assistance. If you do not, then demonstrate the use of either prompt compression techniques or problem-solving and generation of visual outputs.

I have some knowledge of data science and programming. I have done basic python, sql, and tableau in classes I have taken. I feel I struggle with some of the coding. With that being said this completely amazes me. It wrote coding, which I tried and works, in mere seconds. I know I can create this code if asked, but it would have taken me much longer than five minutes. I also know I asked something simple. I think this is just like everything else and the details count. You must know the details of you coding so you can correct it as well as ask the write prompts to get the write coding. That is why I asked it to import the library because that gave it more direction and detail for the way I wanted the coding to go. After I did this I played around with creating coding for a while and it is amazing. I was able to easily create a graph of MLB players’ statistics. I also had it draw roses based on input. It is all amazing how much AI can help with coding.

Prompt-Create python coding using matplotlib that will draw a circle

ChatGPT said:

You can easily draw a circle in Python using matplotlib. Here's a simple example:

```
import matplotlib.pyplot as plt
```

```
# Create a figure and axis
```

```
fig, ax = plt.subplots()
```

```
# Create a circle with center (0,0) and radius 1
```

```
circle = plt.Circle((0, 0), radius=1, color='blue', fill=False, linewidth=2)
```

```
# Add the circle to the axis
```

```
ax.add_patch(circle)
```

```
# Set equal aspect ratio so the circle isn't distorted
```

```
ax.set_aspect('equal', adjustable='box')
```

```
# Set axis limits
```

```
ax.set_xlim(-2, 2)
```

```
ax.set_ylim(-2, 2)
```

```
# Show the plot
```

```
plt.show()
```

This will display a blue circle centered at the origin with radius 1.

☞ Do you want me to also show you how to fill the circle with color or pattern, or just keep it as an outline?

Chapter 4: Prompt Engineering Use Cases in Everyday Tasks

Task: Describe how you would create a prompt (or series of prompts) to automate a task, such as generating a blog post or managing a project using AI-driven prompts.

If I needed to automate a task or generate a blog using prompt engineering, I first need to define what I need to be created. Doing this can be done in many different manners. For blogs you can use a series of layered prompts in order to create a blogpost. I also tried having the AI ask me 20 questions related to writing an article. I thought this worked extremely well. Another everyday scenario is writing emails. I think this was easy and you can create a prompt to work for any situation, you just need to include the details of what you are looking for. I tried a couple different types such as an email, blog, article, list and report. They all came back with adequate results. All of them could use some refining or detail enhancement, but they all would have been acceptable to use as is. Here are the prompts:

“Write an article about 3 Washington Commanders Players in 2025 training camp. Make sure 2 of the players in the article play cornerback and defensive tackle.”

“Write a blog post about a 1 week trip to San Antonio, Texas. Focus on visiting the Alamo and going to a San Antonio Spurs game.”

“Write an email to the Head of Marketing to determine a meeting time and place on September 3rd to discuss marketing strategy with an outside vendor. Make the email sound urgent.”

“Create a list of back-to-school items needed at Loudoun County High School.”

“Generate a report on marketing our new product. Make a list of questions that need answered in order to write the report.”

Chapter 5: Prompt Engineering Use Cases in Everyday Tasks

Task: Use AI to assist in creative writing or research by generating ideas or summarizing information. Include your prompt and what it produced.

Initial Prompt—“Brainstorm 5 plot ideas for a creative writing story that combines sci-fi and fantasy”

ChatGPT said:

Here are 5 fresh plot ideas that blend sci-fi with fantasy:

1. The Starforged Sorcerer

In a far-future galaxy, a dying civilization discovers that their "magic" is actually the manipulation of dark matter through neural implants created by their ancestors. A young exile with raw magical talent must master both spellcraft and quantum mechanics to prevent a war between technologically advanced colonists and the last surviving spellborn.

2. The Archive of Living Worlds

Orbiting an ancient black hole is a library-ship filled with enchanted tomes that contain *entire living universes*. Each book is a pocket reality, half-coded in mathematics, half-woven from spells. When one book begins to bleed into the others, a reluctant apprentice and an AI familiar must travel between realms of dragons, cyborgs, and gods to find the source of the corruption.

3. The Clockwork Dragon of Io

Terraformers on Jupiter’s moon Io awaken something buried in the sulfuric crust: a dragon made of gears, molten core-fire, and circuitry. Worshipped as a god by colonist mystics and hunted as a weapon by the corporate fleet, the dragon bonds with a runaway engineer who realizes its heart is a quantum engine capable of rewriting reality itself.

4. The Last Spell on Earth

Centuries after a solar flare destroyed all advanced technology, Earth lives in a medieval-like age of scattered kingdoms and magic. A mage discovers a dormant satellite AI still orbiting above, whispering data like prophecy. When she uses it, the fusion of spell and code creates the first “techno-sorcery” — and she must decide whether to rebuild a technological society or preserve the fragile magical balance.

5. The Crystal Network

In a galaxy-spanning empire, information travels not through fiber optics but through enchanted crystal conduits—grown, sung into resonance, and powered by starlight. When a hacker-mage uncovers that the

crystals are alive, dreaming, and enslaved, they must rally both digital rebels and mystical orders to free the network, even if it means dismantling the empire's only means of communication.

Chapter 6: Ethics and Considerations in Prompt Engineering

Task: Write a page (about 250 words) discussing the ethical considerations in prompt engineering, including fairness, bias, and data security. Be sure to address all three considerations in your response.

Prompt engineering is the process of shaping how AI systems respond to human input. As these AI systems become more prevalent in our everyday lives, ethical considerations of prompt design become more and more important. There are three main areas when considering ethics in prompt engineering. They are bias mitigation, fairness, and data security.

Fairness means ensuring that prompts do not reinforce stereotypes, exclude underrepresented groups, or over-represent another group. An example is when designing a prompt for a job screening tool, it's essential to avoid language that assumes a particular background or experience. Prompts should be inclusive, accessible, and considerate of diverse perspectives. This is all to ensure that you do not favor one group over another. It basically comes down to me, you need to treat all data equal and not play favorites.

Bias is another huge concern. AI models learn from data, and that data often reflects historical inequalities and prejudices. Even small changes in phrasing can lead to skewed outputs. A prompt like "Why are women less suited for leadership?" reinforces harmful assumptions, while "What challenges do women face in leadership roles?" creates a more balanced response. The goal is for AI to reflect and rate accurately all data without outside prejudices and influence shaping its thought process.

Data security is the third are of ethical considerations. Prompts must never include personal or sensitive information unless explicitly authorized. AI systems are not designed to store or protect private data, so it's your responsibility to anonymize inputs and comply with privacy regulations like GDPR. If you're working in fields like medicine or finance, be especially cautious—accidental data leaks can have serious consequences. People are also very sensitive about their individual data so you do not want to include protected, private information.

Ethical prompt engineering isn't just about avoiding mistakes, it's about building systems that reflect our highest values. By always considering fairness, bias, and security in your work, you'll help create AI tools that are not only intelligent but ethically sound.

Chapter 7: Using ChatGPT Plugins and Custom Instructions

Task: Create a series of at least five-step instructions documenting the process of using ChatGPT plugins and custom instructions to enhance AI interactions.

Step 1: Enable GPT-4 and Access Plugin Features

Select the GPT-4 model in ChatGPT to unlock plugin capabilities. Plugins are not available in GPT-3.5, so this step is essential.

Step 2: Browse and Install Plugins from the Plugin Store

Open the Plugin Store to explore nearly 1,000 plugins across various categories. Click “Install” to add them to your OpenAI account.

Step 3: Activate Plugins for Your Session

You can enable up to three plugins per session from your installed list. Some may require login or authentication with third-party services.

Step 4: Use Prompts That Leverage Plugin Capabilities

Craft specific prompts that match the plugin’s function to trigger its response. For example, ask the Expedia plugin to find flights or the Wolfram plugin to solve equations.

Step 5: Customize ChatGPT’s Behavior with Custom Instructions

Use the “Custom Instructions” feature to personalize how ChatGPT understands and responds to you. This helps tailor tone, depth, and style to your needs.